



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Ms. Mohammed Razia Alangir Banu
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Roll Number	2520090121
Branch	CS-IT
Course Outcome	CO-6

Case Study Topic: TourPlan – Smart Tourism & Travel Itinerary Optimization System using Advanced Non-Linear Data Structures and Application Development

Work Done Summary:

In this case study, advanced non-linear data structures were utilized to develop a smart tourism application called **TourPlan**. The system was designed to manage tourist destinations, travel routes, hotel information, and itinerary planning efficiently. Graphs were used to represent city connectivity, trees were used for destination categorization, and priority-based structures were applied for route recommendations. The implementation demonstrates how non-linear data structures can be used in real-world software applications for tourism management and travel optimization.

Implementation Details:

1. Developed the TourPlan application using non-linear data structures.
2. Represented tourist destinations and city connections using Graphs.
3. Created weighted edges based on travel distance, time, and cost
4. Used Tree structures to organize tourist destinations by category and region.
5. Implemented route recommendation features using graph traversal techniques.
6. Applied shortest-path concepts for efficient travel planning.
7. Managed destination information, hotels, and packages through structured data organization.
8. Generated optimized travel itineraries based on user preferences and budget constraints.

RESULT : SCREENSHOTS/OUTPUT

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Sample Output

===== TOURPLAN SMART TOURISM SYSTEM =====

Available Tourist Destinations:
1. Hyderabad
2. Marangal
3. Vizag
4. Araku
5. Vijayawada

Recommended Tourist Route (DFS Traversal):
Hyderabad Marangal Vizag Araku Vijayawada

Travel Cost : Rs.4500
Travel Duration : 7 Hours

Best Hotels:
1. Hotel Green Valley
2. Tourist Residency

Optimized Itinerary Generated Successfully
CO6 Execution Completed Successfully
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The TourPlan – Smart Tourism & Travel Itinerary Optimization System successfully managed tourist destinations, travel routes, hotel information, and itinerary planning using non-linear data structures. The application efficiently recommended travel paths, optimized route selection, and organized tourism data for better user experience. The results demonstrate the practical implementation of non-linear data structures in developing intelligent tourism and travel management applications.

GitHub link:https://github.com/Jhanasri-thota/DSA_CASE-STUDIES

Student Signature	Faculty Signature